

AKASH MAITY

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EDUCATION

Bachelor of Computer Science

Dalhousie University

2023 : 2026

Halifax, NS

- **Hackathons:** ShiftKey Labs Hackathon: built PSI, a marine digital twin platform for offshore wind siting; Watt the Hack: built a sustainability and green-efficiency web app (React, TypeScript, Supabase)

TECHNICAL SKILLS

- **Languages & Frameworks:** Python, SQL, JavaScript, TypeScript
- **Machine Learning & Deep Learning:** Supervised/unsupervised learning, reinforcement learning (Q-learning, SARSA, DQN, CQL, PPO, Actor-Critic), transfer learning, RNN/GRU/LSTM sequence models, Kalman filtering, recommender systems, ANN semantic search
- **Generative AI & Agents:** Autonomous AI agents, fine-tuning and customizing pre-trained LLMs, RAG (LangChain, FAISS), Hugging Face transformers, NLI-based hallucination detection, prompt engineering, SHAP, LIME, partial dependence plots
- **Data Engineering & Cloud:** ETL design, pipeline orchestration, data preprocessing & feature engineering, automated evaluation/monitoring harnesses, synthetic data generation, AWS, Google Cloud Platform, Firebase, Cloudflare R2, Power BI, REST API integration

PROJECTS

ER Wait-Time Prediction Model, Nova Scotia Hospitals

- Built and deployed a machine learning model to predict emergency-room wait times, achieving an **18.5% reduction** in wait times across Nova Scotia hospitals.

RAG Hallucination Detection over 50 arXiv Reinforcement Learning Papers

- Built a retrieval-augmented generation and faithfulness-evaluation system using FAISS vector search, Hugging Face sentence-transformers, and TinyLlama for generation, with an NLI-based detector that flagged unsupported claims and query expansion that measurably improved retrieval recall.

Multi-Agent Delivery Routing System (VRPTW)

- Designed a multi-agent system with LLM-based intent extraction to convert natural-language delivery requests into structured routing constraints, reaching **100% constraint satisfaction** after iterative prompt optimization, validated against 100 bootstrapped synthetic instances.

Explainable ML Audit, LendingClub Loan Default Prediction

- Audited a LightGBM loan-default model using SHAP, LIME, and partial dependence plots, delivering case narratives and a business-facing slide deck that gave non-technical stakeholders a clear, actionable view of model risk.

Student Personal Assistant, n8n AI Agent

- Built a personal AI agent workflow in n8n that drafts, creates, and sends Gmail messages, tracks a budget, manages Google Calendar events, and answers general and health-related queries end-to-end.

EXPERIENCE

Mobile Application Developer

Collect (Startup Project)

Jan 2026 : May 2026

Halifax, NS

- Led a team building a React Native (Expo) platform for local creators, owning technical decisions across Firebase authentication, Cloudflare R2 media storage, and third-party API integrations.
- Shipped a full-featured application including a video-based project feed, real-time event hub, and persistent light/dark UI.

Teaching Assistant

Dalhousie University

Jan 2024 : Apr 2024

Halifax, NS

- Supported first- and second-year students in a required Computer Science course, mentoring on programming fundamentals and problem solving.

RESEARCH

Research Collaborator, Vessel Trajectory Prediction Pipeline

Ocean Tracking Network (OTN), Dalhousie University (Prof. Gabriel Spadon)

2026 : Present

Halifax, NS

- Build and maintain a full AIS data pipeline for vessel trajectory prediction: parser, sliding-window trajectory batching, spatial filtering, an API ingestor, an evaluation-metrics module, a visual debugger, and a synthetic data generator.
- Implement sequential motion-prediction stages, a Kalman filter baseline followed by a configurable GRU model with a time-decay gate for irregular AIS ping intervals, informed by a literature review of trajectory-forecasting architectures.
- Collaborate with faculty supervisor Prof. Gabriel Spadon via Git-based version control and code review.

LEADERSHIP & EXTRACURRICULAR

Technical Project Lead

Dalhousie Machine Learning Society

May 2026 : Present

Halifax, NS

- Lead end-to-end delivery of applied ML projects for external clients and research partners, defining technical scope and coordinating cross-functional student teams from data acquisition through deployment.
- Designed and delivered a hands-on workshop, "Build Your First AI Agent," using the smolagents framework, and organized the DMLS Summer Hackathon 2026 across five Atlantic Canada industry problem statements.

Student Representative

Dalhousie Machine Learning Society

Nov 2025 : Present

Halifax, NS

- Serve as liaison between the student community and executive leadership; co-organize workshops, hackathons, and mentorship programs.